
CMP6200/DIG6200

Individual Undergraduate Project (FYP)

Project Interim Report

**Improving retail customer's experience
through chatbots using NLP**

Student Name: Usmaan Ahmed

Student ID: 23230300

Course: BSc Cyber Security

Supervisor: Phillip Harris

Date: 12/11/2025

1.0 Introduction and Context

Artificial intelligence has quickly become one of the most revolutionary and impactful developments of recent times. It has benefitted an immense amount of fields, one of them being the retail industry. Through the development of artificial intelligence, natural language processing (NLP) has seen a particular focus due to its unique ability to understand and generate responses closely replicating that of humans. Natural language processing is used in many modern applications, including voice assistants, and chatbots. As the need for communication through digital channels is growing in the retail industry, NLP forms a strong foundation to aid the satisfaction of a customer's experience.

In the UK, the percentage of total online retail sales has increased to 26.5% as of 2022, which is nearly a 15% increase from 2012 (1). It is widely agreed that public image and customer satisfaction are the most important sectors that must be prioritised in order to run and maintain a successful online brand. As the retail industry continues to grow, customer service increasingly plays a crucial role in aiding the maintenance of an online brand, however many retail companies have faced challenges in relation to this. One of the reasons to why they have faced difficulties is due to the use of "vague language" when interacting with customers across omni-channel platforms, including chatbots(2). Customers also express the lack of 24/7 assistance, and the inability to meet their needs of quick and efficient responses(3). Chatbots utilising natural language processing consequently provides a practical solution to meet these demands, providing instant human like responses, whilst being readily available whenever needed.

1.1 Problem Statement

The problem my project is going to address is the matter that retail organisations often fail to meet customer needs in online interactions through their chatbots, resulting in dissatisfaction, hence negatively affecting brand image. This highlights the need for NLP based solutions in order to enhance chatbot performance and improve the overall customer experience.

Review of existing knowledge

Investigating the user experience of customer service chatbot interaction: a framework for qualitative analysis of chatbot dialogues 2021 - Asbjørn Følstad, Cameron Taylor

One report relevant to my study is ‘investigating the user experience of customer service chatbot interaction: a framework for qualitative analysis of chatbot dialogues’ carried out by Asbjørn Følstad and Cameron Taylor. They highlight real word issues faced by customer service chatbots in retail, based on how users interact with them. Their study uses two analytical levels, the turn level and the dialogue level. The turn level examines individual messages, including how relevant the chatbots responses were, and their understandability. The dialogue level looks at broader indicators, including the chatbots helpfulness in its replies, and it’s efficiency.

Findings

Følstad and Taylor found that although chatbots are widely used in the retail industry, their performance often falls short of user expectations. The dialogues they analysed revealed that chatbots frequently misunderstood user inputs, which therefore led to irrelevant or unhelpful responses. Specifically, one of the case studies showed that the 1/3 of the chatbots responses were irrelevant, whilst ¼ of it’s responses were categorised as ‘false positives’, which are responses that were thought to be correct due to the confidence in the way they were delivered, but turned out to be incorrect

The report also noticed that the chatbot was commonly misunderstanding inputs, which therefore led to minimal progression in conversations because of loops of the same responses. This led to the user having to repeatedly rephrase their question. Følstad and Taylor noticed that chatbots were relying on specific wording or phrasing in order to advice the user, meaning that unexpected phrasing, or natural language variation led to misinterpretation and incorrect responses. This is particularly problematic in customer service settings as users will differ in ways they word questions, so if the necessary phrasing isn’t used for the chatbot to recognise, an incorrect response is likely.

Another point raised by this study is that the chatbot had difficulty in providing helpful outcomes. It was found that the suggestion made by the chatbot wasn't always aiding the user to solve their problem, leading them to turning to human agents. This suggests that the chatbot was mainly helpful at broad, surface level interactions, and not as helpful when it came to more in depth and uncommon queries.

Critical Analysis and Gap Identification

One of the limitations of this study is that the authors only method used was analysis of chats between the chatbot and customer. This can be seen as a limitation as the investigation lacked real user ratings and reviews of their experience. This meant that Følstad and Taylor's findings could only consist of their own interpretations, which could have been different to how the user actually felt of their experience. This is a contradiction as the authors claim that analysing dialogue can reveal user frustration and satisfaction or dissatisfaction, however they wouldn't be able to completely verify those feelings without direct contact with the user, and instead left it to their interpretations.

Another limitation of this study is that it is primarily focused on aspects relative to the outcome rather than the actual user experience, which is what the title of the study suggests it is looking into. They analyse the response relevance, understandability, and dialogue progress. These are undeniably important factors when analysing a chatbot, however again they lack analysis of actual user emotions, which is also deemed to be an important consideration of user experiences(2).

Additionally whilst the study does highlight issues such as irrelevant responses, it does not investigate what the underlying reasons could be for those issues. The authors identify that there is a high proportion of false positive responses, but they do not investigate where the errors are stemming from, such as lack of training of their models, or poorly designed intent in it's responses. Not exploring the technical causes of the chatbots limitations and errors leaves a lack of data to improve the chatbot design beyond its basic level of knowledge.

Project Justification

Given that the study uses only inference from dialogue logs without collecting data of user experiences from the user themselves, there is a need for a study that will examine the chatbots conversational performance, and also the perceived quality of those interactions. By developing a chatbot prototype and assessing it's

conversational effectiveness through evaluation of the user's experience, my project will be able to provide a more insightful, and grounded understanding of the customer's experience.

The lack of exploration into the technical causes of chatbot understanding also leaves a gap for my project to fulfil. Limitations in the level of intent recognition and lack of training of models highlights the need for natural language processing techniques which are able to attack these issues. As NLP offers more advanced methods for interpreting user language, with better context and intent, it can be therefore used in order to tackle both these issues through its intent classification features, and its ability to generate contextual responses.

Natural language processing also provides a way to address the lack of investigation into emotional awareness of a chatbot. As emotional intelligence is seen as a significant part of a chatbots ability to provide a high level service(2), NLP covers this with its ability to recognise and respond in accordance with the user's sentiment. Techniques such as sentiment analysis and emotional classification allow it to detect emotions and provide appropriate responses, therefore allowing a chatbot to adapt its tone and provide the relevant reassurance to a user. (4)

3.0 Project Aims, Objectives, and Scope

3.1 Project Aim

The aim of this project is to design, develop and evaluate a NLP based chatbot for a small local business to improve its current customer service by providing accurate, helpful responses and 24/7 availability.

Objective 1: To investigate the business's current method of communication with customers, and gather information from staff to identify challenges and missing aspects that need to be fulfilled by the chatbot by 05/12/25

Objective 2: To collect and analyse common customer queries received by the business, and use them to design the chatbot's intent model and response template by 20/12/25

Objective 3: To develop a functional prototype of a NLP based chatbot using python, RASA and Streamlit to establish the core dialogue and response template by 31/01/26

Objective 4: To customise and enhance the chatbot using NLP techniques using SpaCy, incorporating the relevant customer queries through data collected in objective 2 by 15/03/26

Objective 5: To conduct user testing with customers, collecting feedback and overall satisfaction to identify areas for improvement by 20/04/26

3.3 Project Scope

This project will focus on the design and development of an NLP based chatbot that is specifically designed and tailored to the needs of a small local business. It will include investigating into the business's existing methods of communication and gathering information on customer queries, as well as developing a functional chatbot. The chatbot will be coded using python, and will incorporate NLP techniques in order to customise the chatbot's responses and improve it's ability to address customer queries accurately.

The project will not involve a full release of the chatbot, or integration with a live system such as a social media platform. It will not focus on advanced features such as payment processing or inventory management. The project is limited to creating a functional prototype evaluating its potential to enhance customer service and will not be a production ready system.

4.0 Project Design

4.1 Design and Methods

This project will use development based research supported by user centred evaluation methods. I will use qualitative interviews with the business owner and staff to give me an understanding of the recurring themes, issues and service gaps with the business's current methods of communication.

For objective 2, I will use qualitative content analysis to categorise and structure common customer questions, which will allow me to clearly identify the intents and dialogue templates I need to integrate into the chatbot later on.

For objective 5, I will use user testing in order to evaluate the chatbot's effectiveness. I will use a small group of customers to interact with the system, and give me feedback on their experience.

For objective 6, I will use qualitative thematic analysis to analyse the user feedback and testing results. This will be used to identify the strengths and weaknesses, and opportunities for improvements to the chatbot.

4.2 Justification of Methods

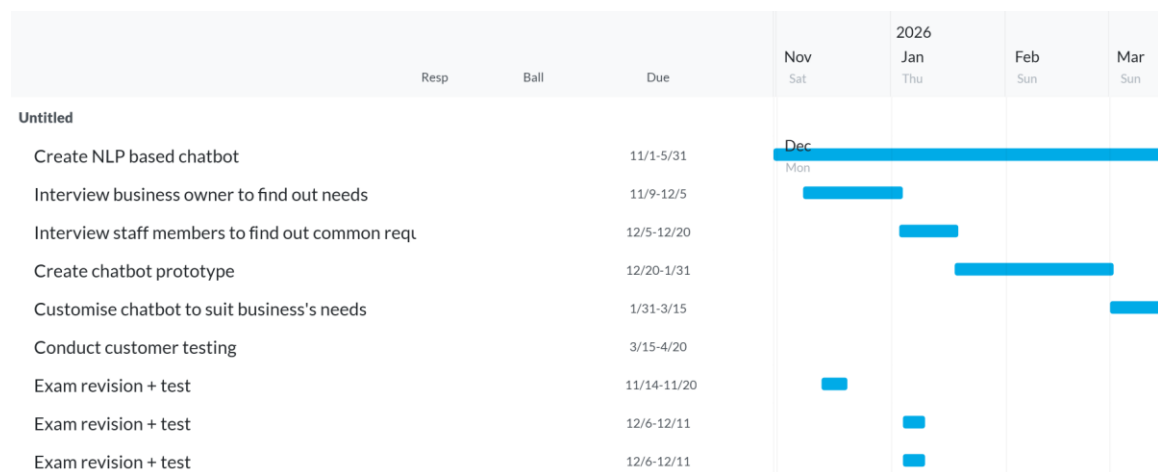
The methods I have selected are appropriate for this project as they directly support the practical and developmental nature of the task. Investigating the business's current communication practices is important in order to provide authentic, and contextual data that cannot be found through literature research alone. I chose

qualitative analysis to do this as the information I am trying to retrieve is likely going to be under an unstructured format, and more conversational which alternative quantitative approaches would not be able to capture.

User testing was also used as the effectiveness of a chatbot cannot be assessed through technical performance alone, and in order to attempt to improve customer satisfaction, user testing is needed to highlight areas done well and areas for improvement. A small group of customers interacting with the prototype provides instant and authentic feedback on usability, accuracy, and overall conversational quality.

Qualitative thematic analysis will be used to interpret the feedback gathered. This method is well suited for identifying the strengths, weaknesses and recurring themes within the user's experience. Thematic analysis allows me to identify why the user responded a certain way, resulting in more meaningful suggestions. Other approaches such as statistical analysis would be less suitable due to the small sample size and qualitative nature of the feedback.

4.3 Project Timeline



5.1 Feasibility

The project is highly feasible within the timeframe, resources available, and skill level required for completion. The scope I have put to together has been made to focus on developing a prototype instead of a fully deployed system. The technical skills required and the tools selected are all open source, and well documented, meaning there is a wide range of resources I can use to aid my work, and there is no financial investment needed to access these applications or resources.

From a skills perspective, this project aligns within my capabilities, and the deadlines set are fair and achievable, therefore I am confident in my ability to carry out this project.

5.2 Risk Analysis and Mitigation

Potential Risk	Likelihood (Low/Med/High)	Impact (Low/Med/High)	Mitigation Strategy
<i>Limited user participation during testing</i>	<i>Low</i>	<i>High</i>	<i>Ask the owner to invite regular customers, or use the current customer interaction channels to invite customers in.</i>
<i>Insufficient customer query data</i>	<i>Medium</i>	<i>High</i>	<i>Request to interview other staff, preferably members who are constantly in contact with customers through their current channels.</i>

Table 4.1: Risk Assessment and Mitigation

5.3 Ethical Considerations

As the project will involve engagement and feedback from human participants, ethical considerations are a central component of the projects design. When interviewing the owner and staff, there will be informed consent obtained before any discussions are made. The participants will be fully aware of the purpose of the project, and their participation will be completely voluntary with the right to withdraw at any time without consequence. As well as this, no personal identifiers of the customers will be recorded, and all participation of the testing will be strictly in the moment feedback that I will note down anonymously. The chatbot will not have access to any past or ongoing customer details, or order details. The chatbot will focus on queries regarding products and services offered by the business.

Any participants in the user testing will be given a brief overview of the chatbot before they interact with it. They will be assured that their feedback is confidential and is solely for the use of improving the chatbot and will not be published anywhere.

References

- (1)Centre for retail research. (n.d) *Online trends & statistics for UK, Europe & North America 2022-2024*. <https://www.retailresearch.org/online-retail.html>
- (2) Holmes, T. L., & Brewer, E. C. (2020). *Customer service challenges in omni-channel retailing — An exploratory study of vague language in retailer customer service policies*.

Atlantic Marketing Journal.

https://www.researchgate.net/publication/344458133_Customer_Service_Challenges_in_Omni-Channel_Retailing-

[An_Exploratory_Study_of_Vague_Language_in_Retailer_Customer_Service_Policies](#)

(3) Valente, C. (2024, June 5). *Current customer expectations in the retail industry.*

GoContact. <https://www.gocontact.com/blog/contact-center-2/current-customer-expectations-in-the-retail-industry/>

(4) Følstad, A. & Taylor, C. (2021). *Investigating the user experience of customer service chatbot interaction: a framework for qualitative analysis of chatbot dialogues.*

Quality and User Experience, 6, 6, <https://doi.org/10.1007/s41233-021-00046-5>